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Transcendental Logarithmic Utility Functions

By LAURITS R. CHRISTENSEN, DALE W. JORGENSON, AND LAWRENCE J. LAU*

The traditional starting point for econometric studies of consumer demand is a system of demand functions giving the quantity consumed of each commodity as a function of total expenditure and the prices of all commodities. Tests of the theory of demand are formulated by requiring that the demand functions be consistent with utility maximization. Additive and homothetic utility functions have played an important role in formulating tests of the theory of demand. If the utility function is homothetic, expenditure proportions are independent of total expenditure. If the utility function is additive and homothetic, elasticities of substitution among all pairs of commodities are constant and equal.¹

An example of the traditional approach to demand analysis is the system of double logarithmic demand functions employed in the pioneering studies of consumer demand by Henry Schultz, Richard Stone, and Herman Wold. If the theory of demand is valid and demand functions are double logarithmic, the utility function is linear logarithmic.² Similarly, the Rotterdam system of demand functions employed by A. P. Barten and Henri Theil is consistent with utility maximization only if the utility function is linear logarithmic.³ A linear logarithmic utility function is both additive and homothetic; all expendi-

ture proportions are constant, and elasticities of substitution among all pairs of commodities are constant and equal to unity.

Hendrik Houthakker and Stone have developed alternative approaches to demand analysis that retain the assumption of additivity while dropping the assumption of homotheticity.⁴ Stone has employed a linear expenditure system, based on a utility function that is linear in the logarithms of quantity consumed less a constant for each commodity. The constants are interpreted as initial commitments; incremental expenditure proportions, derived from quantities consumed in excess of the initial commitments, are constant for all variations in total expenditure and prices. If all initial commitments are zero, the utility function is linear logarithmic in form. Non-zero commitments permit expenditure proportions to vary with total expenditure.

Houthakker has employed a direct addilog system, based on a utility function that is additive in functions that are homogeneous in the quantity consumed for each commodity. The degree of homogeneity may differ from commodity to commodity, permitting expenditure proportions to vary with total expenditure. If the degree of homogeneity is the same for all commodities, the addilog utility function is additive and homothetic. Robert Basmann, Leif Johansen, and Kazuo Sato have combined the approaches of Houthakker and Stone, defining each of the homogeneous functions in the direct addilog

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¹ The class of additive and homothetic utility functions was first characterized by Abram Bergson.

² See Schultz, Stone (1954a), Wold, and Robert Basmann, R. C. Battalio, and J. H. Kagel.

³ See Barten (1964, 1967, 1969), Daniel McFadden, and Theil (1965, 1967, 1971).

⁴ See Houthakker (1960) and Stone (1954b). The linear expenditure system was originally proposed by Lawrence Klein and Herman Rubin.

utility function on the quantity consumed less a constant for each commodity. The resulting utility function is additive but not homothetic.⁵

Our first objective is to develop tests of the theory of demand that do not employ additivity or homotheticity as part of the maintained hypothesis. For this purpose we introduce new representations of the utility function in Section I. Our approach is to represent the utility function by functions that are quadratic in the logarithms of the quantities consumed. The resulting utility functions provide a local second-order approximation to any utility function. These utility functions allow expenditure shares to vary with the level of total expenditure and permit a greater variety of substitution patterns among commodities than functions based on constant and equal elasticities of substitution among all pairs of commodities.

Our second objective is to exploit the duality between prices and quantities in the theory of demand. A complete model of consumer demand implies the existence of an indirect utility function, defined on total expenditure and the prices of all commodities.⁶ The indirect utility function is homogeneous of degree zero and can be expressed as a function of the ratios of prices of all commodities to total expenditure. The indirect utility function is useful in characterizing systems of direct demand functions, giving quantities consumed as functions of the ratios of prices to total expenditure. The direct utility function is useful in characterizing systems of indirect demand functions, giving the ratios of prices to total expenditure as functions of the quantities consumed. The system consisting of direct utility function and in-

direct demand functions is dual to the system consisting of indirect utility function and direct demand functions.⁷

We represent the indirect utility function by functions that are quadratic in the logarithms of ratios of prices to total expenditure, paralleling our treatment of the direct utility function. The resulting indirect utility functions provide a local second-order approximation to any indirect utility function. These indirect utility functions are not required to be additive or homothetic. The duality between direct and indirect utility functions has been used extensively in Houthakker's pathbreaking studies of consumer demand. Paralleling the direct addilog demand system, Houthakker has employed an indirect addilog system, based on an indirect utility function that is additive in ratios of prices to total expenditure.⁸

We refer to our representation of the direct utility function as the direct transcendental logarithmic utility function, or more simply, the *direct translog utility function*. The utility function is a transcendental function of the logarithms of quantities consumed. Similarly we refer to our representation of the indirect utility function as the indirect transcendental logarithmic utility function, or, more simply, the *indirect translog utility function*. Earlier, we introduced transcendental logarithmic functions into the study of production.⁹ The duality between direct and indirect translog utility functions is analogous to the duality between translog pro-

⁵ A recent survey of econometric studies of consumer demand is given by Alan Brown and Angus Deaton.

⁶ The indirect utility function was introduced by Giovanni Antonelli, and independently by Harold Hotelling.

⁷ Indirect demand functions were introduced by Antonelli. The duality between direct and indirect utility functions is discussed by Lau (1969a); the duality between systems of direct and indirect demand functions is discussed by Wold, John Chipman, and Leonid Hurwicz.

⁸ See Houthakker (1960). This demand system was originally proposed by Conrad Leser and has also been employed by W. H. Somermeyer, J. G. M. Hilhorst, and J. W. W. A. Wit.

⁹ See the authors (1971, 1973).

duction and price frontiers employed in our study of production.

For an additive direct utility function ratios of indirect demand functions, giving the ratios of prices, depend only on the quantities of the two commodities involved. The direct addilog and linear expenditure systems, together with the system employed by Basmann, Johansen, and Sato, have this property. Similarly, for an additive indirect utility function, ratios of direct demand functions giving the ratios of quantities depend only on the prices of the two commodities involved. The indirect addilog system has this property. For an additive and homothetic direct utility function the ratios of indirect demand functions depend only on the ratios of quantities. Furthermore, the indirect utility function is also additive and homothetic, so that ratios of direct demand functions depend only on ratios of prices. The use of direct and indirect translog utility functions permits us to test these restrictions on direct and indirect demand functions. We do not impose the restrictions as part of the maintained hypothesis.

We present statistical tests of the theory of demand in Section II. These tests can be divided into two groups. First, we test restrictions on the parameters of the direct translog utility function implied by the theory of demand. We test these restrictions without imposing the assumptions of additivity and homotheticity. We test precisely analogous restrictions on the parameters of the indirect translog utility function. Second, we test restrictions on the direct translog utility function corresponding to restrictions on the form of the utility function. In particular, we test restrictions corresponding to additivity and homotheticity of the direct translog utility function. Again, we test precisely analogous restrictions on the indirect translog utility function.

We present empirical tests of the theory

of demand based on time-series data for the United States for 1929–72 in Section III. The data include prices and quantities consumed of the services of consumers' durables, nondurable goods, and other services. For these data we present direct tests of the theory of demand based on the direct translog utility function, and indirect tests of the theory based on the indirect translog utility function. For both direct and indirect tests we first test the extensive set of restrictions implied by the theory of demand. Proceeding conditionally on the validity of the theory, we test restrictions on the form of the direct and indirect utility functions implied by the assumptions of additivity and homotheticity.

I. Transcendental Logarithmic Utility Functions

A. Direct Translog Utility Function

The *direct utility function* U can be represented in the form:

$$(1) \quad \ln U = \ln U(X_1, X_2, \dots, X_m)$$

where X_i is the quantity consumed of the i th commodity. The consumer maximizes utility subject to the budget constraint:

$$(2) \quad \sum p_i X_i = M$$

where p_i is the price of the i th commodity and M is the value of total expenditure.

The first-order conditions for a maximum of utility can be written:

$$(3) \quad \frac{\partial \ln U}{\partial \ln X_j} = \mu \frac{p_j X_j}{U} \quad (j = 1, 2, \dots, m)$$

where μ is the marginal utility of income. From the budget constraint we obtain:

$$(4) \quad \frac{\mu}{U} = \frac{1}{M} \sum \frac{\partial \ln U}{\partial \ln X_i}$$

so that:

$$(5) \quad \frac{\partial \ln U}{\partial \ln X_j} = \frac{p_j X_j}{M} \sum \frac{\partial \ln U}{\partial \ln X_i} \quad (j = 1, 2, \dots, m)$$

To preserve symmetry with our treatment of the indirect utility function given below, we approximate the negative of the logarithm of the direct utility function by a function quadratic in the logarithms of the quantities consumed:

$$(6) \quad -\ln U = \alpha_0 + \sum \alpha_i \ln X_i + \frac{1}{2} \sum \sum \beta_{ij} \ln X_i \ln X_j$$

Using this form for the utility function we obtain:

$$(7) \quad \alpha_j + \sum \beta_{ji} \ln X_i = \frac{p_j X_j}{M} \sum (\alpha_k + \sum \beta_{ki} \ln X_i) \quad (j = 1, 2, \dots, m)$$

To simplify notation we write:

$$(8) \quad \alpha_M = \sum \alpha_k$$

$$(9) \quad \beta_{Mi} = \sum \beta_{ki} \quad (i = 1, 2, \dots, m)$$

so that:

$$(10) \quad \frac{p_j X_j}{M} = \frac{\alpha_j + \sum \beta_{ji} \ln X_i}{\alpha_M + \sum \beta_{Mi} \ln X_i} \quad (j = 1, 2, \dots, m)$$

The budget constraint implies that:

$$(11) \quad \sum \frac{p_i X_i}{M} = 1$$

so that, given the parameters of any $m-1$ equations for the budget shares $p_j X_j / M$ ($j=1, 2, \dots, m$), the parameters of the m th equation, α_m and β_{mj} ($j=1, 2, \dots, m$), can be determined from the definitions of α_M and β_{Mi} ($j=1, 2, \dots, m$).

Since the equations for the budget shares are homogeneous of degree zero in the parameters, a normalization of the parameters is required for estimation. A convenient normalization for the parameters of the direct translog utility function is:

$$(12) \quad \alpha_M = \sum \alpha_i = -1$$

B. Indirect Translog Utility Function

The indirect utility function V can be represented in the form:

$$(13) \quad \ln V = \ln V \left(\frac{p_1}{M}, \frac{p_2}{M}, \dots, \frac{p_m}{M} \right)$$

We determine the budget share for the j th commodity from the identity:¹⁰

$$(14) \quad \frac{p_j X_j}{M} = - \frac{\partial \ln V}{\partial \ln p_j} \bigg/ \frac{\partial \ln V}{\partial \ln M} \quad (j = 1, 2, \dots, m)$$

Preserving symmetry with the direct utility function, we approximate the logarithm of the indirect utility function by a function quadratic in the logarithms of the ratios of prices to the value of total expenditure:

$$(15) \quad \ln V = \alpha_0 + \sum \alpha_i \ln \frac{p_i}{M} + \frac{1}{2} \sum \sum \beta_{ij} \ln \frac{p_i}{M} \ln \frac{p_j}{M}$$

Using this form for the utility function we obtain:

$$(16) \quad \frac{\partial \ln V}{\partial \ln p_j} = \alpha_j + \sum \beta_{ji} \ln \frac{p_i}{M} \quad (j = 1, 2, \dots, m)$$

$$(17) \quad - \frac{\partial \ln V}{\partial \ln M} = \sum \left(\alpha_k + \sum \beta_{ki} \ln \frac{p_i}{M} \right)$$

As before, we simplify notation by writing:

$$(18) \quad \alpha_M = \sum \alpha_k$$

$$(19) \quad \beta_{Mi} = \sum \beta_{ki} \quad (i = 1, 2, \dots, m)$$

so that:

$$(20) \quad \frac{p_j X_j}{M} = \frac{\alpha_j + \sum \beta_{ji} \ln \frac{p_i}{M}}{\alpha_M + \sum \beta_{Mi} \ln \frac{p_i}{M}} \quad (j = 1, 2, \dots, m)$$

¹⁰ This is the logarithmic form of René Roy's Identity.

The budget constraint implies that given the parameters of any $m-1$ equations for the budget shares, the parameters of the m th equation, α_m and β_{mj} ($j=1, 2, \dots, m$), can be determined from the definitions of α_M and β_{Mj} ($j=1, 2, \dots, m$). As before, we can normalize the parameters so that:

$$(21) \quad \alpha_M = \sum \alpha_i = -1$$

II. Testing the Theory of Demand

A. Stochastic Specification

The first step in implementing an econometric model of demand based on the direct translog utility function is to add a stochastic specification to the theoretical model based on equations for the budget shares $p_j X_j / M$ ($j=1, 2, \dots, m$). Given the disturbances in any $m-1$ equations, the disturbance of the remaining equation can be determined from the budget constraint. Only $m-1$ equations are required for a complete econometric model of demand.

B. Equality and Symmetry

We estimate $m-1$ equations for the budget shares, subject to normalization of the parameter α_M appearing in each equation at minus unity. If the equations are generated by utility maximization, the parameters β_{mj} ($j=1, 2, \dots, m$) appearing in each equation must be the same. This results in a set of restrictions relating the m parameters appearing in each of the $m-1$ equations, a total of $m(m-2)$ restrictions. We refer to these as *equality restrictions*.

The logarithm of the direct translog utility function is twice differentiable in the logarithms of the quantities consumed, so that the Hessian of this function is symmetric. This gives rise to a set of restrictions relating the parameters of the cross-partial derivatives:

$$(22) \quad \beta_{ij} = \beta_{ji} \quad (i \neq j; i, j = 1, 2, \dots, m)$$

There are $(1/2)(m-2)(m-1)$ restrictions of this type among the parameters of the $m-1$ equations we estimate directly and $m-1$ such restrictions among the parameters of the equations we estimate indirectly from the budget constraint. We refer to these as *symmetry restrictions*. The total number of symmetry restrictions is $(1/2)m(m-1)$.

If equations for the budget shares are generated by maximization of a direct translog utility function, the parameters satisfy equality and symmetry restrictions. There are $(1/2)m(3m-5)$ such restrictions.

C. Additivity

If the direct utility function U is *additive*, we can write

$$(23) \quad \ln U = F(\sum \ln U^i(X_i))$$

where each of the functions U^i depends on only one of the commodities demanded X_i , and F is a real-valued function of one variable.

The parameters of the translog approximation to an additive direct utility function can be written:

$$(24) \quad -\frac{\partial \ln U}{\partial \ln X_i} = -F' \frac{\partial \ln U^i}{\partial \ln X_i} = \alpha_i \quad (i = 1, 2, \dots, m)$$

$$(25) \quad -\frac{\partial^2 \ln U}{\partial \ln X_i \partial \ln X_j} = -F'' \frac{\partial \ln U^i}{\partial \ln X_i} \frac{\partial \ln U^j}{\partial \ln X_j} = \beta_{ij} \quad (k \neq j; i, j = 1, 2, \dots, m)$$

where the logarithmic derivatives,

$$(26) \quad F' = \frac{\partial \ln U}{\partial \ln U^i} \quad (i = 1, 2, \dots, m)$$

$$(27) \quad F'' = \frac{\partial^2 \ln U}{\partial \ln U^i \partial \ln U^j} \quad (i, j = 1, 2, \dots, m)$$

are independent of i and j .

Under additivity the parameters of the translog utility function satisfy the restrictions:

$$(28) \quad \beta_{ij} = \theta \alpha_i \alpha_j \\ (i \neq j; i, j = 1, 2, \dots, m)$$

where

$$(29) \quad \theta = - \frac{F''}{(F')^2}$$

We refer to these as *additivity restrictions*. The total number of such restrictions is $(1/2)(m-2)(m-1)$.

The translog approximation to an additive utility function is not necessarily additive. The direct translog utility function is additive if and only if $\ln U$ can be written as the sum of m functions, each depending on only one of the quantities demanded. We refer to such a function as *explicitly additive*. Explicit additivity of the translog utility function implies the additivity restrictions given above and the additional restriction:

$$(30) \quad \theta = 0$$

We refer to this restriction as the *explicit additivity restriction*. We note that the translog approximation to an explicitly additive function is explicitly additive.

D. Homotheticity

If the direct utility function is *homothetic*, we can write:

$$(31) \quad \ln U = F(\ln H(X_1, X_2, \dots, X_m))$$

where the function H is homogeneous of degree one.

The parameters of the translog approximation to a homothetic direct utility function can be written:

$$(32) \quad - \frac{\partial \ln U}{\partial \ln X_i} = \frac{\partial F}{\partial \ln H} \frac{\partial \ln H}{\partial \ln X_i} = \alpha_i \\ (i = 1, 2, \dots, m)$$

$$(33) \quad - \frac{\partial^2 \ln U}{\partial \ln X_i \partial \ln X_j} = \\ \left[\frac{\partial F}{\partial \ln H} \frac{\partial^2 \ln H}{\partial \ln X_i \partial \ln X_j} + \frac{\partial^2 F}{\partial \ln H^2} \frac{\partial \ln H}{\partial \ln X_i} \frac{\partial \ln H}{\partial \ln X_j} \right] = \beta_{ij} \\ (i, j = 1, 2, \dots, m)$$

Homogeneity of degree one of the function H implies that:

$$(34) \quad \sum \frac{\partial \ln H}{\partial \ln X_i} = 1$$

$$(35) \quad \sum_{j=1}^m \frac{\partial^2 \ln H}{\partial \ln X_i \partial \ln X_j} = 0 \\ (i = 1, 2, \dots, m)$$

Under homotheticity the parameters of the translog utility function satisfy the restrictions:

$$(36) \quad \beta_{Mi} = \sigma \alpha_i \quad (i = 1, 2, \dots, m)$$

where

$$(37) \quad \sigma = \frac{\partial^2 F}{\partial \ln H^2}$$

and we have used the normalization

$$(38) \quad \sum \alpha_i = -1$$

We refer to these as *homotheticity restrictions*. There are $m-1$ such restrictions.

The translog approximation to a homothetic utility function is not necessarily homothetic. A necessary and sufficient condition for homotheticity of the translog utility function is that it is homogeneous, so that:

$$(39) \quad \sigma = 0$$

We refer to this restriction as the *homogeneity restriction*. We note that the translog approximation to a homogeneous function is homogeneous.

E. Additivity and Homotheticity

The class of additive and homothetic direct utility functions coincides with the class of utility functions with constant elasticities of substitution among all commodities. If the direct utility function U is additive and homothetic, we can write:

$$(40) \quad \ln U = F(\sum \delta_i X_i^\rho)$$

or

$$(41) \quad \ln U = F(\sum \delta_i \ln X_i)$$

The second form is a limiting case of the first corresponding to unitary elasticities of substitution among all commodities.

The parameters of the translog approximation to an additive and homothetic utility function satisfy the additivity and homotheticity restrictions given above, so that:

$$(42) \quad \beta_{ij} = \theta \alpha_i \alpha_j \quad (i \neq j; i, j = 1, 2, \dots, m)$$

$$(43) \quad \beta_{ii} = (\sigma + \theta) \alpha_i + \theta \alpha_i^2 \\ (i = 1, 2, \dots, m)$$

where we have used the normalization

$$(44) \quad \sum \alpha_i = -1$$

The translog approximation to an additive and homothetic utility function is not necessarily additive and homothetic. We can, however, identify the parameters of a translog approximation with the parameters of the additive and homothetic utility function given above, as follows:

$$(45) \quad \sigma + \theta = \rho$$

$$(46) \quad \alpha_i = \delta_i \quad (i = 1, 2, \dots, m)$$

As before, the parameter θ corresponds to:

$$(47) \quad \theta = - \frac{F''}{(F')^2}$$

The translog approximation to a utility function characterized by unitary elasticities of substitution among all com-

modities satisfies the restriction:

$$(48) \quad \sigma + \theta = \rho = 0$$

Under this restriction the expenditures shares are constant:

$$(49) \quad \frac{p_j X_j}{M} = \frac{\alpha_j + \theta \alpha_j \sum \alpha_i \ln X_i}{-1 - \theta \sum \alpha_i \ln X_i} = -\alpha_j$$

The value of the parameter θ is arbitrary so that we can let σ and θ be equal to zero. The translog approximation to a utility function with unitary elasticities of substitution has the same empirical implications as a linear logarithmic utility function, which is explicitly additive and homogeneous.

F. Duality

In implementing an econometric model of demand based on the indirect translog utility function the first step as before is to add a stochastic specification to the theoretical model based on equations for the budget shares $p_j X_j / M$ ($j = 1, 2, \dots, m$). Only $m-1$ equations are required for a complete model. If equations for the budget shares are generated from the indirect translog utility function, the parameters satisfy equality and symmetry restrictions that are strictly analogous to the corresponding restrictions for the direct translog utility function.

Additivity and homotheticity restrictions for the indirect translog utility function are analogous to the corresponding restrictions for the direct translog function. The direct utility function is homothetic if and only if the indirect utility function is homothetic.¹¹ In general, an additive direct utility function does not correspond to an additive indirect utility function.¹² However, if the direct utility function is additive and homothetic, the indirect util-

¹¹ See, for instance, Paul Samuelson (1965) and Lau (1969b).

¹² See Lau (1969b).

ity function is additive and homothetic. Direct and indirect utility functions corresponding to the same preferences are additive only if the direct utility function is homothetic¹³ or the direct utility function is linear logarithmic in all but one of the commodities.¹⁴ In this relationship the roles of direct and indirect utility functions can be interchanged.

A direct utility function is *self-dual* if the corresponding indirect utility function has the same form.¹⁵ The only direct translog utility function which is self-dual is the linear logarithmic utility function. Linear logarithmic utility functions are the only additive and homothetic direct or indirect translog utility functions. Translog direct and indirect utility functions represent the same preferences if and only if they are self-dual; that is, if and only if they are linear logarithmic. Unless this stringent condition is met, the direct and indirect translog approximations to a given pair of direct and indirect utility functions represent different preferences.

III. Empirical Results

A. Summary of Tests

We have developed statistical tests of the theory of demand that do not employ the assumptions of additivity and homotheticity for individual commodities or for commodity groups. At this point we outline restrictions on equations for the budget shares corresponding to direct and indirect translog utility functions. First, we present restrictions derived from the theory of demand; that is, from the basic hypothesis of utility maximization. Second,

we present restrictions derived from hypotheses about functional form such as additivity and homotheticity.

Our empirical tests are based on data for three commodity groups—services of consumers' durables, nondurable goods, and other services—so that we specialize our presentation of statistical tests of the theory of demand to the three-commodity case, $m=3$. A complete econometric model for either direct or indirect translog utility functions is provided by any pair of equations for the budget shares. We consider the system of two equations,

$$(50) \quad \frac{p_1 X_1}{M} = \frac{\alpha_1 + \beta_{11} \ln X_1 + \beta_{12} \ln X_2 + \beta_{13} \ln X_3}{-1 + \beta_{M1} \ln X_1 + \beta_{M2} \ln X_2 + \beta_{M3} \ln X_3}$$

$$(51) \quad \frac{p_2 X_2}{M} = \frac{\alpha_2 + \beta_{21} \ln X_1 + \beta_{22} \ln X_2 + \beta_{23} \ln X_3}{-1 + \beta_{M1} \ln X_1 + \beta_{M2} \ln X_2 + \beta_{M3} \ln X_3}$$

corresponding to the direct translog utility function and the system of equations,

$$(52) \quad \frac{p_1 X_1}{M} = \frac{\alpha_1 + \beta_{11} \ln \frac{p_1}{M} + \beta_{12} \ln \frac{p_2}{M} + \beta_{13} \ln \frac{p_3}{M}}{-1 + \beta_{M1} \ln \frac{p_1}{M} + \beta_{M2} \ln \frac{p_2}{M} + \beta_{M3} \ln \frac{p_3}{M}}$$

$$(53) \quad \frac{p_2 X_2}{M} = \frac{\alpha_2 + \beta_{21} \ln \frac{p_1}{M} + \beta_{22} \ln \frac{p_2}{M} + \beta_{23} \ln \frac{p_3}{M}}{-1 + \beta_{M1} \ln \frac{p_1}{M} + \beta_{M2} \ln \frac{p_2}{M} + \beta_{M3} \ln \frac{p_3}{M}}$$

corresponding to the indirect translog utility function. Restrictions for the two systems are perfectly analogous so that the

¹³ See Houthakker (1960), Samuelson (1965), and Lau (1969b).

¹⁴ This is the special case introduced by John Hicks. See also Samuelson (1969).

¹⁵ See Samuelson (1965) and Houthakker (1965). We may also mention the "self-dual addilog system" introduced by Houthakker (1965). This system is not generated by additive utility functions except for special cases.

following outline may be applied to either system. We emphasize the fact that the two systems represent the same preferences if and only if

$$(54) \quad \beta_{ij} = 0 \quad (i, j = 1, 2, 3)$$

1. *Equality Restrictions.* The parameters $\{\beta_{M1}, \beta_{M2}, \beta_{M3}\}$ occur in both equations and must take the same values. There are three equality restrictions.

2. *Symmetry Restrictions.* One restriction of this type is explicit in the two equations we estimate directly, namely,

$$(55) \quad \beta_{12} = \beta_{21}$$

In addition, we estimate the parameters β_{31} and β_{32} from the equations:

$$(56) \quad \beta_{31} = \beta_{M1} - \beta_{11} - \beta_{21}$$

$$(57) \quad \beta_{32} = \beta_{M2} - \beta_{12} - \beta_{22}$$

so that the two additional symmetry restrictions are implicit in the two equations we estimate. We write these restrictions in the form:

$$(58) \quad \beta_{13} = \beta_{M1} - \beta_{11} - \beta_{21}$$

$$(59) \quad \beta_{23} = \beta_{M2} - \beta_{12} - \beta_{22}$$

There are three symmetry restrictions altogether.

We can identify tests of the six equality and symmetry restrictions with tests of the theory of demand. We next consider tests of hypotheses about functional form.

3. *Additivity Restrictions.* Given the equality and symmetry restrictions, the additivity restrictions take the form:

$$(60) \quad \beta_{12} = \theta \alpha_1 \alpha_2$$

$$(61) \quad \beta_{13} = \theta \alpha_1 (-1 - \alpha_1 - \alpha_2)$$

$$(62) \quad \beta_{23} = \theta \alpha_2 (-1 - \alpha_1 - \alpha_2)$$

We introduce the additional parameter θ , so that there are two independent restrictions of this type.

4. *Homotheticity Restrictions.* Given the equality and symmetry restrictions, the homotheticity restrictions take the form:

$$(63) \quad \beta_{M1} = \sigma \alpha_1$$

$$(64) \quad \beta_{M2} = \sigma \alpha_2$$

$$(65) \quad \beta_{M3} = \sigma (-1 - \alpha_1 - \alpha_2)$$

We introduce the additional parameter σ , so that there are two independent restrictions of this type.

5. *Explicit Additivity Restrictions.* Given the equality, symmetry, and additivity restrictions, a translog utility function is explicitly additive under the further restriction:

$$(66) \quad \theta = 0$$

6. *Homogeneity Restrictions.* Given the equality, symmetry, and homotheticity restrictions, the homogeneity restriction takes the form:

$$(67) \quad \sigma = 0$$

7. *Linear Logarithmic Utility Restrictions.* Given the equality, symmetry, additivity, and homotheticity restrictions, the translog utility function reduces to linear logarithmic form under the additional restrictions:

$$(68) \quad \sigma = \theta = 0$$

Tests of hypotheses about the form of the utility function can be carried out in a number of different sequences. We propose to test additivity and homotheticity in parallel. Each of these hypotheses consists of two independent restrictions. Our test procedure is presented in diagrammatic form in Figure 1. We also present alternative test procedures. For example, we could first test additivity and then test homotheticity only if additivity is accepted. Our test procedure is indicated with double lines while possible alternative procedures are indicated with single lines.

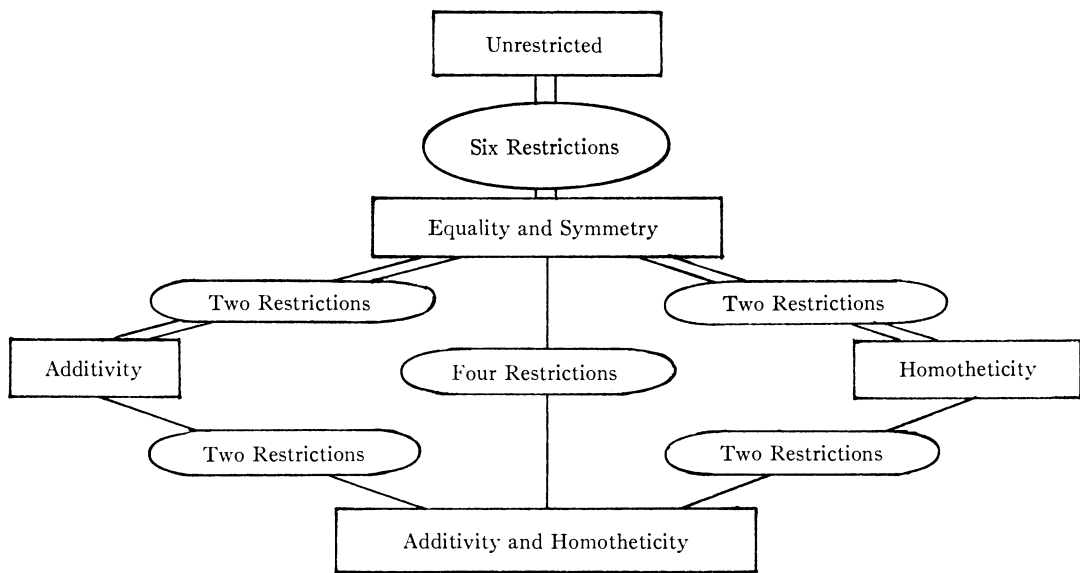


FIGURE 1. TESTS OF ADDITIVITY AND HOMOTHETICITY

The next step in our test procedure depends on the outcome of the tests of additivity and homotheticity. If we accept both additivity and homotheticity restrictions, we continue by testing explicit additivity and homogeneity in parallel. If we accept both hypotheses, we conclude that

utility is linear logarithmic. A second possibility is that we accept additivity but reject homotheticity. In this case we continue by testing explicit additivity. The third possibility is that we accept homotheticity but reject additivity; we continue by testing homogeneity. If we reject both

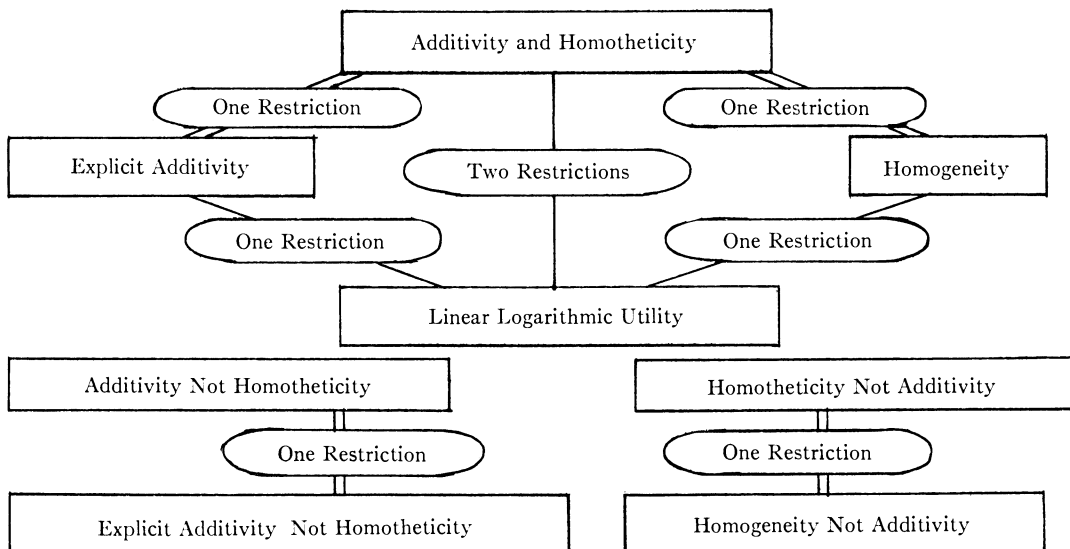


FIGURE 2. TESTS CONDITIONAL ON ADDITIVITY OR HOMOTHETICITY

additivity and homogeneity, we terminate the sequence of tests. These tests are presented diagrammatically in Figure 2.

B. *Estimation*

Our empirical results are based on time-series data for *U. S.* personal consumption expenditures for 1929–72. The data include prices and quantities of the services of consumers' durables, nondurable goods, and other services. We have fitted the equations for budget shares generated by direct translog and indirect translog utility functions, employing the stochastic specification outlined above. Under this specification only two equations are required for a complete econometric model of demand. We have fitted equations for the services of consumers' durables (durables) and for nondurable goods (nondurables). There are forty-four observations for each behavioral equation, so that the number of degrees of freedom available for statistical tests of the theory of demand is eighty-eight for either direct or indirect specification.

For both direct and indirect specifications the maintained hypothesis consists of the unrestricted form of the two behavioral equations for the budget shares. We estimate the behavioral equations for durables and nondurables by the method of maximum likelihood and derive estimates of the parameters of the behavioral equation for services, using the budget constraint.¹⁶ The maximum likelihood estimates of the parameters of all three behavioral equations are invariant with respect to the choice of the two equations to be estimated directly.

¹⁶ We employ the maximum likelihood estimator discussed, for example, by Edmond Malinvaud, pp. 338–41. In the computations we use the Gauss-Newton method described by Malinvaud, p. 343. For the direct series of tests we assume that the disturbances are independent of the quantities consumed. For the indirect series of tests we assume that the disturbances are independent of the ratios of prices to the value of total expenditure.

The unrestricted behavioral equations, normalized so that the parameter α_M is minus unity in each equation, involve fourteen unknown parameters or seven unknown parameters for each equation. Unrestricted estimates of these parameters for the direct translog utility function are presented in the first column of Table 1. The first hypothesis to be tested is that the theory of demand is valid; to test this hypothesis we impose the six equality and symmetry restrictions. Restricted estimates of the fourteen unknown parameters are presented in the second column of Table 1. Unrestricted and restricted estimates of the fourteen unknown parameters for the indirect translog utility function are presented in the first and second columns of Table 2.

Given the validity of the theory of demand, the remaining hypotheses to be tested are restrictions on the functional form. First, we test the hypothesis that the direct utility function is additive; this hypothesis requires that we impose two additional restrictions. The corresponding restricted estimates of the unknown parameters for the direct translog utility function are given in the third column of Table 1. Second, we test the hypothesis that the direct utility function is homothetic without imposing the additivity restrictions; this hypothesis requires two restrictions in addition to the equality and symmetry restrictions. The restricted estimates of the unknown parameters for the direct translog utility function are given in the fourth column of Table 1. Finally, we impose both additivity and homotheticity restrictions, obtaining the restricted estimates presented in the fifth column of Table 1. The corresponding restricted estimates for the indirect translog utility function are given in the third, fourth, and fifth columns of Table 2.

The next stage of our test procedure is contingent on the outcome of our tests of

TABLE 1—ESTIMATES OF THE PARAMETERS OF THE DIRECT TRANSLOG UTILITY FUNCTION

Parameters	Unrestricted Estimates	Equality and Symmetry	Additivity	Homotheticity	Additivity and Homotheticity	Explicit Additivity, Not Homotheticity	Homogeneity, Not Additivity	Explicit Additivity and Homogeneity	Additivity and Homogeneity	Linear Logarithmic Utility
DURABLES										
α_1	-.145 (.00302)	-.138 (.00262)	-.144 (.00357)	-.139 (.00257)	-.146 (.00328)	-.147 (.00279)	-.140 (.00220)	-.147 (.00265)	-.147 (.00280)	-.127 (.00378)
β_{11}	.0490 (.0888)	-.0216 (.0129)	-.137 (.00971)	-.268 (.0140)	-.123 (.00778)	-.102 (.00694)	-.317 (.0122)	-.100 (.00637)	-.121	—
β_{12}	.500 (.194)	.139 (.0277)	.142 (.0227)	.170 (.0293)	.0983 (.0219)	—	.147 (.0178)	—	.0671 (.00386)	—
β_{13}	-.555 (.206)	-.0737	.115	-.139	.0790	—	.170	—	.0543	—
β_{M1}	.659 (.864)	.0440 (.0356)	.120	.0407 (.0372)	.0542	-.102	—	-.100	—	—
β_{M2}	3.53 (1.76)	.165 (.113)	.213	.136	.176	-.339	—	-.323	—	—
β_{M3}	-3.94 (1.96)	.265 (.107)	.209 (.104)	.116	.141	-.275 (.0318)	—	-.261	—	—
NONDURABLES										
α_2	-.467 (.00420)	-.468 (.00353)	-.472 (.00345)	-.464 (.00402)	-.473 (.00385)	-.471 (.00487)	-.461 (.00328)	-.472 (.00440)	-.471 (.00336)	-.495 (.00792)
β_{21}	.272 (.299)	.139 (.0277)	.142 (.0227)	.170 (.0293)	.0983 (.0219)	—	.147 (.0178)	—	.0671 (.00386)	—
β_{22}	-.536 (.774)	-.334 (.0563)	-.306 (.0571)	-.168 (.0619)	-.179	-.339 (.0448)	-.216 (.0326)	-.323	-.241	—
β_{23}	.259 (.866)	.361	.377	.133	.256	—	.0693	—	.174	—
β_{M1}	.282 (.555)	.0440 (.0356)	.120	.0407 (.0372)	.0542	-.102	—	-.100	—	—
β_{M2}	-.271 (1.55)	.165 (.113)	.213	.136	.176	-.339	—	-.323	—	—
β_{M3}	.105 (1.69)	.265 (.107)	.209 (.104)	.116	.141	-.275 (.0318)	—	-.261	—	—

additivity and homotheticity. If we accept additivity, but not homotheticity, we test explicit additivity by imposing one further restriction. If we accept homotheticity, but not additivity, we test homogeneity by imposing one further restriction. The restricted estimates under these restrictions are given in the sixth and seventh columns of Table 1. The corresponding estimates for the indirect translog utility function are given in the sixth and seventh columns of Table 2.

If we accept both additivity and homotheticity hypotheses, we test the hypothesis that the direct utility function is explicitly additive by imposing one further restriction. Similarly, we test homogeneity by imposing one further restriction. Finally, we impose both restrictions under the hypothesis that utility is linear logarithmic. The restricted estimates under each of these sets of restrictions are given

in the eighth through tenth columns of Table 1. The corresponding restricted estimates for the indirect translog utility function are given in the eighth through tenth columns of Table 2. The direct and indirect translog utility functions are self-dual if they are linear logarithmic, so that restricted estimates for the two alternative econometric models given in the final columns of the two tables are identical.

C. Test Statistics

To test the validity of restrictions implied by the theory of demand and restrictions on the form of the utility function, we employ test statistics based on the likelihood ratio λ , where:

$$(69) \quad \lambda = \frac{\max_{\omega} \mathcal{L}}{\max_{\Omega} \mathcal{L}}$$

TABLE 2—ESTIMATES OF THE PARAMETERS OF THE INDIRECT TRANSLOG UTILITY FUNCTION

Parameters	Unrestricted Estimates	Equality and Symmetry	Additivity	Homotheticity	Additivity and Homotheticity	Explicit Additivity, Not Homotheticity	Homogeneity, Not Additivity	Explicit Additivity and Homotheticity	Additivity and Homogeneity	Linear Logarithmic Utility
DURABLES										
α_1	-.141 (.00215)	-.125 (.00300)	-.128 (.00279)	-.119 (.00390)	-.119 (.00370)	-.127 (.00228)	-.120 (.00392)	-.120 (.00364)	-.120 (.00370)	-.127 (.00378)
β_{11}	-.115 (.0188)	-.0970 (.0188)	-.0595 (.00972)	-.131 (.0273)	-.0930 (.00823)	-.0662 (.00684)	-.106 (.0246)	-.105 (.00618)	-.0796	—
β_{12}	.695 (.0447)	.0816 (.0300)	.0188 (.0153)	.100 (.0403)	.0282 (.0131)	—	.113 (.0379)	—	.0454 (.00238)	—
β_{13}	-.473 (.0527)	-.0174	.0144	-.0334	.0212	—	-.00754	—	.0342	—
β_{M1}	-.801 (.172)	-.0324 (.0379)	-.0262	-.0644 (.0216)	-.0434	-.0662	—	-.105	—	—
β_{M2}	4.52 (.419)	-.489 (.134)	-.491	-.273	-.183	-.631	—	-.440	—	—
β_{M3}	-3.50 (.464)	-.280 (.114)	-.274 (.107)	-.202	-.137	-.393 (.213)	—	-.330	—	—
NONDURABLES										
α_2	-.472 (.00472)	-.499 (.00533)	-.494 (.00479)	-.506 (.0576)	-.503 (.00597)	-.497 (.00383)	-.504 (.00612)	-.503 (.00576)	-.502 (.00612)	-.495 (.00792)
β_{21}	-.872 (.120)	.0816 (.0300)	.0188 (.0153)	.100 (.0403)	.0282 (.0131)	—	.113 (.0379)	—	.0454 (.00238)	—
β_{22}	1.11 (.247)	-.668 (.0881)	-.565 (.0715)	-.539 (.0810)	-.300	-.631 (.0307)	-.348 (.0610)	-.440	-.189	—
β_{23}	-.227 (.164)	.0972	.0560	.166	.0891	—	.234	—	.143	—
β_{M1}	-1.63 (.212)	-.0324 (.0379)	-.0262	-.0644 (.0216)	-.0434	-.0662	—	-.105	—	—
β_{M2}	2.98 (.464)	-.489 (.134)	-.491	-.273	-.183	-.631	—	-.440	—	—
β_{M3}	-1.11 (.334)	-.280 (.114)	-.274 (.107)	-.202	-.137	-.393 (.213)	—	-.330	—	—

The likelihood ratio is the ratio of the maximum value of the likelihood function $\mathcal{L}$ for the econometric model of demand ω subject to restriction to the maximum value of the likelihood function for the model Ω without restriction. For normally distributed disturbances the likelihood ratio is equal to the ratio of the determinant of the restricted estimator of the variance-covariance matrix of the disturbances to the determinant of the unrestricted estimator, each raised to the power $-(n/2)$.

Our test statistic for each set of restrictions is based on minus twice the logarithm of the likelihood ratio, or:

$$(70) \quad -2\ln\lambda = n(\ln |\hat{\Sigma}_\omega| - \ln |\hat{\Sigma}_\Omega|)$$

where $\hat{\Sigma}_\omega$ is the restricted estimator of the variance-covariance matrix and $\hat{\Sigma}_\Omega$ is the unrestricted estimator. Under the null hypothesis this test statistic is distributed,

asymptotically, as chi-square with number of degrees of freedom equal to the number of restrictions to be tested.

To control the overall level of significance for each series of tests, direct and indirect, we set the level of significance for each series at .05. We allocate the overall level of significance among the various stages in each series. We first assign a level of significance of .01 to the test of equality and symmetry restrictions implied by the theory of demand. We then assign a level of significance of .04 to tests of restrictions on functional form. These two sets of tests are "nested"; under the null hypothesis the sum of levels of significance provides a close approximation to the level of significance for both sets of tests simultaneously.

We test additivity and homotheticity in parallel, proceeding conditionally on the validity of the theory of demand. These tests are not nested so that the sum of

levels of significance for each of the two hypotheses is an upper bound for the level of significance of tests of the two hypotheses considered simultaneously. There are four possible outcomes of our parallel tests of additivity and homotheticity: Reject both, accept both, accept only additivity, and accept only homotheticity. If we reject both hypotheses, our test procedure terminates. If we accept either or both of these hypotheses, we proceed to test explicit additivity or homogeneity, or both.

If we accept both additivity and homotheticity, we test explicit additivity and homogeneity in parallel, conditional on the validity of additivity and homotheticity. Again, the tests are not nested, so that the sum of levels of significance for the two tests provides an upper bound for the tests considered simultaneously. If we accept only additivity, we proceed to test explicit additivity, conditional on additivity. Similarly, if we accept only homotheticity, we proceed to test homogeneity.

Since our three alternative procedures for testing explicit additivity and homogeneity are mutually exclusive, we carry out tests of these hypotheses one time at most. We assign levels of significance to the four hypotheses—additivity, explicit additivity, homotheticity, and homogeneity—with the assurance that the sum of levels of significance provides an upper bound for the level of significance for all four tests. We assign a level of significance of .01 to each of the four hypotheses. With the aid

of critical values for our test statistics given in Table 3 the reader can evaluate the results of our tests for alternative allocations of the overall levels of significance among stages of the test procedure.

Test statistics for both direct and indirect tests of the theory of demand and of restrictions on functional form are presented in Tables 4 and 5. At a level of significance of .01 we reject the hypothesis that restrictions implied by the theory of demand are valid for either direct or indirect series of tests. With this conclusion we can terminate the test sequence, since these results invalidate the theory of demand.

One interesting alternative to our test procedure is to maintain the theory of demand and to test the validity of restrictions on the form of the utility function. Proceeding conditionally on the validity of the theory of demand, we could test the validity of restrictions on the form of the utility function. For the direct series of tests we would reject the restrictions implied by additivity and homotheticity. This conclusion would hold for our preferred procedure of testing these hypotheses in parallel, for a test of either hypothesis conditional on the validity of the

TABLE 3—CRITICAL VALUES OF χ^2 /DEGREES OF FREEDOM

Degrees of Freedom	Level of Significance				
	.10	.05	.025	.01	.005
1	2.71	3.84	5.02	6.63	7.88
2	2.33	3.00	3.69	4.61	5.30
4	1.94	2.37	2.79	3.32	3.72
6	1.77	2.10	2.41	2.80	3.09

TABLE 4—TEST STATISTICS FOR DIRECT AND INDIRECT TESTS OF THE THEORY OF DEMAND AND TESTS OF ADDITIVITY AND HOMOTHETICITY

Hypothesis	Degrees of Freedom	Direct	Indirect
	<u>Theory of Demand</u>		
Equality and Symmetry	6	5.64	10.25
<u>Given the Theory of Demand</u>			
Additivity	2	33.63	2.95
Homotheticity	2	13.71	22.68
Additivity and Homotheticity	4	19.02	16.24
Additivity Given Homotheticity	2	24.33	9.81
Homotheticity Given Additivity	2	4.42	29.54

other, or for a joint test of the two hypotheses.

For the indirect series of tests we would reject the restrictions implied by homotheticity, but we would accept the restrictions implied by additivity at a level of significance of .01. For the direct series of tests, we could terminate the test sequence conditional on the theory of demand at this point. For the indirect series of tests, we could test the validity of explicit additivity, given additivity but not homotheticity. We would accept the hypothesis of explicit additivity, given additivity at a level of significance of .01, conditional on the validity of the theory of demand.

In previous econometric studies of demand, the theory of demand has been maintained together with the assumption of additivity. A second alternative to our test procedure is to maintain both the theory of demand and the restrictions implied by additivity. We can test homo-

theticity; for either direct or indirect series of tests we reject this hypothesis. Given additivity but not homotheticity, we would reject explicit additivity for the direct series of tests and accept explicit additivity for the indirect series of tests. Either of these sets of results would rule out linear logarithmic utility.

IV. Summary and Conclusion

Our objective has been to test the theory of demand without imposing the assumptions of additivity and homotheticity as part of the maintained hypothesis. For either the direct or the indirect series of tests, we conclude that the theory of demand is inconsistent with the evidence. These results confirm the findings of Wold (in association with Lars Jüréen) for the double logarithmic demand system and Barten for the Rotterdam demand system.¹⁷ At the same time our results provide the basis for more specific conclusions.

If the theory of demand were valid, the double logarithmic form for the system of demand functions would imply that the utility function is linear logarithmic. Similarly, the validity of the theory of demand and the Rotterdam form for the system of demand functions would imply linear logarithmic utility. An equally valid interpretation of the results of Wold and Barten is that the theory of demand is valid, but that utility is not linear logarithmic. Our results rule out this alternative interpretation and make possible an unambiguous rejection of the theory of demand.

A possible alternative to our test procedure is to maintain the validity of the theory of demand. Proceeding conditionally on the validity of demand theory, we would reject the hypothesis of additivity for the direct series of tests. Additivity of the direct utility function is employed as a

TABLE 5—TEST STATISTICS FOR DIRECT AND INDIRECT TESTS OF RESTRICTIONS ON FUNCTIONAL FORM, GIVEN ADDITIVITY OR HOMOTHETICITY OR BOTH

Hypothesis	Degrees of Freedom	Direct	Indirect
<u>Given Additivity</u>			
Explicit Additivity	1	19.97	1.59
<u>Given Homotheticity</u>			
Homogeneity	1	1.47	4.73
<u>Given Additivity and Homotheticity</u>			
Explicit Additivity	1	11.25	4.02
Homogeneity	1	1.74	1.16
Linear Logarithmic Utility	2	29.67	48.31
<u>Given Explicit Additivity and Homotheticity</u>			
Linear Logarithmic Utility	1	48.09	92.61
<u>Given Additivity and Homogeneity</u>			
Linear Logarithmic Utility	1	57.60	95.46

¹⁷ See Wold, especially pp. 281–302; Barten (1969). A detailed review of tests of the theory of demand is given by Brown and Deaton, pp. 1188–95.

maintained hypothesis by Houthakker in the direct addilog demand system, by Stone in the linear expenditure system, and by Basmann, Johansen, and Sato in the demand system incorporating features of both.

Again, proceeding conditionally on the validity of the theory of demand, we would accept the hypothesis of additivity for the indirect series of tests. This hypothesis is maintained by Houthakker in the implementation of the indirect addilog demand system. Additivity of the direct and indirect utility functions is mutually consistent if and only if the direct utility function is homothetic, so that the acceptance of additivity of the indirect utility function and the rejection of homotheticity rule out acceptance of additivity for the direct utility function.

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